

TEZPUR UNIVERSITY
Mid Term Test, Autumn 2016
C0 504 Natural Language Processing

Marks : 40

Time : 60 minutes

The figures in the right-hand margin indicate marks for the individual questions.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

Submit question no. 8 within first fifteen minutes.

1. What are some applications of natural language processing?
Point out how different such applications require different levels of language processing capabilities. 3 + 2
2. Write a Perl script that prints those lines of input that begin with *The* or *This*. 3
3. How are *n-gram* knowledge and *Markov assumption* used in estimating the probability of a sequence of words? 4
4. Consider the following sentences-

The classes will remain suspended for seven days at a stretch next week. Many students are planning to go home. They will be with their families during Durga Puja.

- (a) Suggest a Parts-of-speech (PoS) tagset that will be required to do PoS tagging of the above sentences. 3
- (b) Perform PoS tagging for the above sentences. 3

5. (a) Write a context free grammar (CFG) that can recognise simple imperative sentences of English containing noun phrases and other constituents. 3
- (b) Give an example of an imperative sentence of English. Using the CFG proposed for the above question draw the parse tree for the sentence. 3
6. How is Earley's parser more suitable for natural languages compared to conventional parsers used by compilers? 3
7. Suppose in a sentence construct there is a noun and a verb, and the number, person and gender of these two words have to match. How can features and unification be used to check the validity of a sentence. 2

Roll No. _____

Detach this page and submit it within first fifteen minutes.

8. (I) Fill in the blanks 1×6
- (a) The smallest meaning bearing unit in a natural language expression is a _____ (word / morpheme / syllable).
 - (b) Witten-Bell discounting is a _____ (morphological / smoothing) technique.
 - (c) If a corpus has N words, then the number of tri-grams is _____.
 - (d) The parts-of-speech of the word *beautifully* is _____ (noun / verb / adjective / adverb).
 - (e) The phrase *beautiful morning breeze* is _____ (incorrect / ambiguous).
 - (f) A verb that requires a direct object is a _____ (transitive / intrinsic / nonfinite) verb.
- (II) State whether each of the following statements are true or false- 1×5
- (a) For Roman characters ASCII values are same as Unicode values.
 - (b) A given corpus can provide better statistical information about *higher* order n-grams.
 - (c) In general an compiled program will run faster than a corresponding interpreted program.
 - (d) Earley's parser does not require prior parts-of-speech disambiguation.
 - (e) If an the value of an attribute in a feature structure is not specified, it will lead to failure of unification of that feature structure with another.